WORKED SOLUTION

STEP I 2007, Question 12

Probability and Statistics • Conditional Probability

STEP I 2007, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q12

The key idea

For the second sweet we don't know what the bags look like, because that depends on what happened first. So we condition: split according to everything that happened before the second draw, work out the chance of red in each case, and add up using the law of total probability. In part (ii) there's a useful shortcut: the expected number of red sweets in a bag that has just received a random sweet is easy to write down, so we needn't split on the colour of the transferred sweet more than once.

Solution

Part (i)

Let R_1 be the event that the first sweet is red and R_2 the event that the second sweet is red. Every sweet is equally likely to be drawn first, so

$$\mathrm{P}(R_1) = \frac{a}{N}.$$

For R_2, condition on the colour of the first sweet. If the first is red, $N-1$ sweets remain, of which $a-1$ are red; if it is not red, $N-1$ remain, of which a are red. (Here $N \geq 2$ guarantees that a second sweet exists.) By the law of total probability,

$$\begin{aligned}\mathrm{P}(R_2) &= \mathrm{P}(R_1)\,\mathrm{P}(R_2 \mid R_1) + \mathrm{P}(R_1')\,\mathrm{P}(R_2 \mid R_1') \\ &= \frac{a}{N}\cdot\frac{a-1}{N-1} + \frac{N-a}{N}\cdot\frac{a}{N-1} \\ &= \frac{a[(a-1)+(N-a)]}{N(N-1)} = \frac{a(N-1)}{N(N-1)} = \frac{a}{N}.\end{aligned}$$

So $\mathrm{P}(R_2) = \mathrm{P}(R_1)$, as required. (The same is true by symmetry: every ordered pair of distinct sweets is equally likely, so the second sweet is just as likely to be any particular sweet as the first is.)

Part (ii)

Here the "first sweet" is the one transferred after the first toss, and the "second sweet" is the one chosen after the second toss. Let R_1, R_2 be the events that these are red. The two tosses are independent, and each is independent of which sweets are chosen.

The first sweet. Conditioning on the first toss,

$$\mathrm{P}(R_1) = p \cdot \frac{a}{N} + q \cdot \frac{b}{N} = \frac{pa + qb}{N}.$$

The second sweet. There are four cases for the two tosses.

Heads, Heads. A sweet is taken from bag 1, and then another sweet is taken from bag 1. That is exactly the situation of part (i) applied to bag 1 (with N sweets, a red), so the second sweet is red with probability a/N.

Heads, Tails. A sweet moves from bag 1 to bag 2, which now has $N+1$ sweets; then we draw from bag 2. Bag 2 has $b+1$ red sweets with probability a/N and b red sweets with probability $1 - a/N$, so

$$\mathrm{P}(R_2 \mid \mathrm{HT}) = \frac{a}{N} \cdot \frac{b+1}{N+1} + \left(1 - \frac{a}{N}\right) \frac{b}{N+1} = \frac{b + \frac{a}{N}}{N+1} = \frac{Nb + a}{N(N+1)}.$$

? **Ask yourself** Why does the second line collapse so neatly?

The numerator is the expected number of red sweets in bag 2 after the transfer: the b it started with, plus the chance a/N that the incoming sweet is red. Since bag 2 has a fixed size $N+1$ in this case, the chance of drawing red is that expected number divided by $N+1$.

Tails, Heads. By the same argument with the bags swapped, bag 1 now holds $N+1$ sweets with an expected $a + b/N$ red ones, so

$$\mathrm{P}(R_2 \mid \mathrm{TH}) = \frac{Na + b}{N(N+1)}.$$

Tails, Tails. Part (i) applied to bag 2 gives probability b/N.

Now combine the four cases, with probabilities p^2, pq, qp, q^2:

$$\begin{aligned}
\mathrm{P}(R_2) &= p^2 \frac{a}{N} + pq \frac{(Nb + a) + (Na + b)}{N(N+1)} + q^2 \frac{b}{N} \\
&= p^2 \frac{a}{N} + pq \frac{(N+1)(a+b)}{N(N+1)} + q^2 \frac{b}{N} \\
&= \frac{p^2 a + pqa + pqb + q^2 b}{N} \\
&= \frac{pa(p+q) + qb(p+q)}{N} = \frac{pa + qb}{N},
\end{aligned}$$

using $p + q = 1$ in the last step. This equals $\mathrm{P}(R_1)$, which is what we had to show.

Checks

- If $p = 1$, everything happens in bag 1, and the answer a/N for both sweets is exactly part (i). If $p = 0$ we get part (i) for bag 2.
- If $a = b = 0$ there are no red sweets and both probabilities are 0; if $a = b = N$ every sweet is red and both are 1.
- With $N = 5$, $a = 2$, $b = 4$, $p = 0.3$ the formula gives $(0.6 + 2.8)/5 = 0.68$. A simulation of 200 000 runs of the whole process gave 0.679 for the first sweet and 0.681 for the second, and exact enumeration over all small N, a, b agrees.

Common slips

- Forgetting that after a transfer the receiving bag has $N + 1$ sweets, not N, and the giving bag has $N - 1$.
- In the HH and TT cases, treating the second draw as independent of the first. It isn't, but part (i) tells us the answer is unchanged anyway, so use it rather than redoing the calculation.

Beyond STEP

Both parts are instances of a deep idea: if you draw from an urn without looking at what you took, your knowledge of the urn hasn't changed, so the next draw has the same distribution. Formally, the sequence of draws is *exchangeable*, and in part (ii) the quantity $p \times (\text{proportion red in bag } 1) + q \times (\text{proportion red in bag } 2)$, the chance that the next sweet drawn is red, is a *martingale*: a quantity whose expected future value equals its present value. Martingales are central to modern probability, from gambling to finance. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet conditional expectation and these ideas.

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